From Structure to Surprisal: Hierarchical Syntax and Prediction Difficulty in Transformer Language Models

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Abstract

Human language exhibits strong structural regularities that are often explained through cognitive memory constraints, including Dependency Length Minimization (DLM) and increasing processing difficulty under deeper syntactic embedding. This project investigates whether transformer-based language models exhibit similar behavior. We combine dependency length analysis on human and LLM generated corpora with controlled sentence experiments measuring surprisal, perplexity, and entropy under increasing structural complexity. We further compare original and surface modified sentences to examine whether observed effects arise from deep syntax or shallow cues. Our results show that LLM generated text often displays short dependency structures resembling DLM, but increasing dependency length does not consistently increase prediction difficulty. Hierarchical embedding affects surprisal only partially, while reducing surface cues makes structural effects more visible. These findings suggest that transformer models capture some human-like linguistic patterns, but rely heavily on surface level regularities rather than robust hierarchical representations.

1. MOTIVATION FOR THE RESEARCH PROBLEM

Human language is shaped by memory limitations. One of the most widely discussed consequences of this is *Dependency Length Minimization* (DLM), the tendency for syntactic dependencies to remain relatively short. Another closely related idea is that increasing structural complexity, such as center embedding, makes sentence processing harder for humans. These observations raise an important question: do transformer-based language models show similar structural tendencies and processing difficulty patterns, or do they succeed mainly through statistical pattern matching?

This project studies that question from two complementary directions. First, we compare dependency length patterns in human written and LLM-generated text. Second, we examine whether increasing syntactic hierarchy makes prediction harder for a language model, measured through surprisal, perplexity, and entropy. Together, these analyses help evaluate whether LLM behavior reflects genuinely human-like structural constraints or only surface-level regularities.

Research Question: Do transformer based language models exhibit human like structural constraints, specifically Dependency Length Minimization and increased prediction difficulty under greater syntactic hierarchy?

Objective: To evaluate whether increasing syntactic complexity, measured through dependency length and embedding depth, leads to increased prediction difficulty in LLMs, and to compare these patterns with human language.

2. RELATED WORK

Prior work in psycholinguistics suggests that human sentence structure is shaped by memory constraints, often reflected in shorter dependencies and reduced processing load. Dependency Length Minimization has been widely studied as a principle explaining why human languages prefer structures that reduce long-distance syntactic relationships. In parallel, surprisal theory proposes that processing difficulty is linked to the unpredictability of a word given its preceding context.

In NLP, transformer-based language models have achieved strong predictive performance, but their degree of syntactic understanding remains debated. Some studies suggest that these models can capture agreement, constituency, and long-range dependencies, while other work shows that they often rely on shallow heuristics or lexical and positional cues. This motivates a direct comparison between structural properties of human language and LLM behavior under controlled syntactic manipulations.

Our work brings these lines together by combining dependency-based structural analysis, controlled embedding experiments, and prediction-based metrics to examine whether LLMs behave in a human-like way under structural complexity.

3. BACKGROUND

Dependency Length. In dependency grammar, a sentence is represented as a tree of head-dependent relations. The dependency length of a relation is the linear distance between the head and the dependent in the sentence. Shorter average dependency lengths are often associated with reduced cognitive load.

Surprisal. Surprisal measures how unexpected a token is in context. A token with low probability under the model has high surprisal, and therefore greater prediction difficulty.

Center Embedding. Center embedding refers to nested syntactic structures such as relative clauses inserted inside other clauses. These constructions are known to increase processing difficulty in humans and are useful for testing hierarchical sensitivity in language models.

4. HYPOTHESES AND PREDICTIONS

- **Hypothesis 1: Dependency Length Minimization in LLMs.** LLM-generated text will prefer shorter dependency structures. **Prediction:** Average dependency length in LLM text will be broadly comparable to human language.
- **Hypothesis 2: Dependency Length vs. Prediction Difficulty.** Longer dependencies increase prediction difficulty. **Prediction:** Higher dependency length will be associated with higher surprisal and perplexity.
- **Hypothesis 3: Effect of Hierarchical Embedding.** Increasing syntactic depth increases prediction difficulty. **Prediction:** Greater embedding depth will increase surprisal at key tokens such as the main verb.
- **Hypothesis 4: Role of Surface Cues.** Surface patterns influence model behavior strongly. **Prediction:** Reducing surface cues will make the effect of structural complexity more visible.

5. METHODS

5.1 Data

We used four types of data to analyze structure and prediction difficulty:

1. **Human Language Data:** Sentences from Universal Dependencies Treebanks for English, French, Hindi, Marathi, and Spanish. These provide a baseline for human syntactic structure.
2. **LLM-Generated Text:** A large generated-text dataset containing approximately **25,000 sentences,** used for dependency-length statistics.

3. **Controlled Sentences:** Artificially generated around **30000** sentences with progressively increasing structural complexity, including simple sentences, modifier expansions, and center embedded clauses.

4. **Modified Surface Data:** Variants of controlled sentences with reduced surface cues, used to test whether model behavior changes when shallow signals are weakened.

5.2 Dependency Length Analysis

Dependency structures were extracted using a dependency parser. For each sentence, dependency length was computed as:

$$DL = \sum |position(head) - position(dependent)| \tag{1}$$

Using this, we computed:

- Average Dependency Length (ADL)
- Maximum Dependency Length
- Dependency Length Distribution
- Tree Height

This analysis was performed on both human language datasets and LLM-generated text to compare structural tendencies across the two sources.

5.3 Prediction-Based Metrics

To measure prediction difficulty, we used a pretrained transformer language model (GPT-2). For each sentence, we computed token-level and sentence-level quantities.

Surprisal:

$$S(w_t) = -\log P(w_t \mid w_{<t}) \tag{2}$$

Entropy:

$$H_t = -\sum_{w \in V} P(w \mid w_{<t}) \log P(w \mid w_{<t}) \tag{3}$$

Perplexity:

$$PP = \exp\left(\frac{1}{N}\sum_{t=1}^{N} S(w_t)\right) \tag{4}$$

These metrics were computed for controlled sentences and their modified variants. The main focus was whether prediction difficulty increases with dependency length and hierarchical depth.

5.4 Experimental Design

We conducted three main experiments:

Experiment 1: Human vs. LLM Structural Comparison. Dependency statistics were computed on human language corpora and LLM-generated text to test whether model-generated language exhibits DLM-like patterns.

Experiment 2: Dependency Length vs. Prediction Difficulty. Controlled sentences with increasing structural complexity were used to examine whether longer dependencies increase surprisal and perplexity.

Experiment 3: Hierarchical Embedding and Surface Form. Center-embedded sentences and their surface-modified versions were tested to determine whether prediction difficulty reflects deep structure or shallow surface cues.

Comparison : Human Surprisal Approximation While direct human surprisal measurements (e.g., reading time or eye-tracking data) are not collected in this study, prior work in psycholinguistics has established that human processing difficulty correlates with surprisal, where less predictable words require greater processing effort. Based on this, we approximate human surprisal trends using structural complexity measures such as dependency length and embedding depth.

This allows us to qualitatively compare human prediction difficulty with LLM-based surprisal, while acknowledging that human surprisal is indirectly estimated rather than directly measured.

5.5 Multi-Model and Attention-Based Analysis

To further analyze model behavior, we extend our experiments to multiple transformer language models, including GPT-2, DistilGPT-2, and GPT-2 Medium. For each model, we compute average surprisal and maximum token surprisal across sentences with increasing embedding depth. This allows us to evaluate whether prediction difficulty patterns are consistent across architectures.

In addition, we perform attention-based analysis using GPT-2 to examine how the model distributes attention across tokens. We extract

attention weights from the final layer and analyze both token-level attention patterns (via heatmaps) and aggregate trends of attention to earlier context as sentence complexity increases.

5.6 Inference and Analysis

To evaluate the hypotheses, we applied:

- **Trend Analysis:** observing how surprisal, entropy, and perplexity change with dependency length and embedding depth
- **Correlation Analysis:** measuring the relationship between dependency length and surprisal, and between embedding depth and surprisal
- **Comparative Analysis:** comparing human vs. LLM dependency distributions and original vs. modified sentence behavior

5.7 Implementation Scale

The project covers five human languages using UD Treebanks and approximately **55,000+** LLM generated sentences. We produced more than 20 visual analyses across dependency, surprisal, and embedding experiments. In addition, we developed an interactive **web interface** that allows users to input sentences and observe model behavior in real time, including token-level surprisal and top-k predictions.

6. RESULTS

6.1 Dependency Length in LLMs

The analysis of LLM-generated text shows that average dependency length remains relatively low, with distributions skewed toward shorter dependencies. This suggests that LLM-generated text exhibits a pattern broadly consistent with Dependency Length Minimization. The observed average dependency length is approximately 3.6, which falls within a range comparable to natural-language baselines. The dependency length analysis on human language datasets shows a similar trend, with distributions strongly skewed toward shorter dependencies across all five languages. The average dependency length remains low and consistent, reflecting the well-established principle of Dependency Length

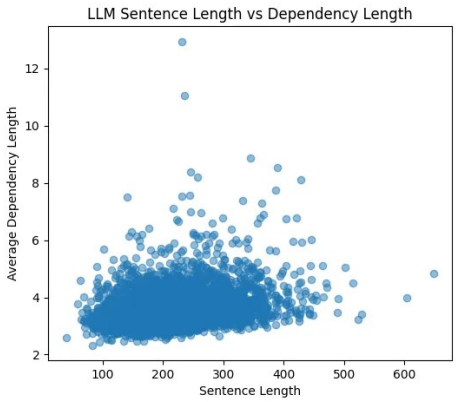

Figure 1: LLM Analysis

Minimization in natural language. Compared to LLM-generated text, human data exhibits slightly tighter distributions with fewer long-distance dependencies, indicating stronger structural constraints. This provides a reliable baseline for evaluating whether LLMs replicate human-like dependency patterns.

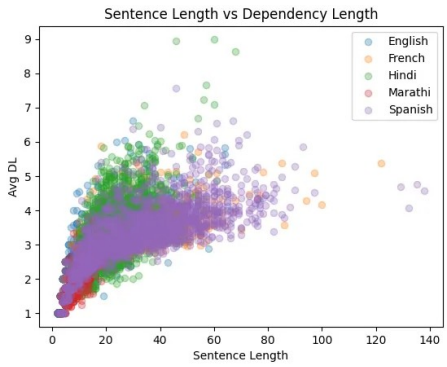

Figure 2: Human text analysis

6.2 Human vs LLM Structural Comparison

- Human language shows a strong positive relationship between sentence length and dependency length (correlation = 0.72), indicating structured and consistent growth in syntactic complexity.
- LLM-generated text exhibits a much weaker relationship (correlation = 0.25), suggesting that dependency structure is less tightly coupled with sentence length.
- The growth rate of dependency length is significantly higher in human language (0.047) compared to LLM-generated

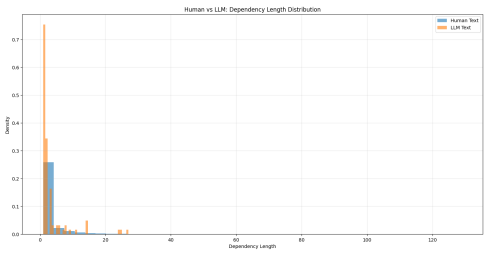

Figure 3: Human V/s LLM structure density comparison

text (0.002), indicating that human sentences scale structural complexity more systematically.

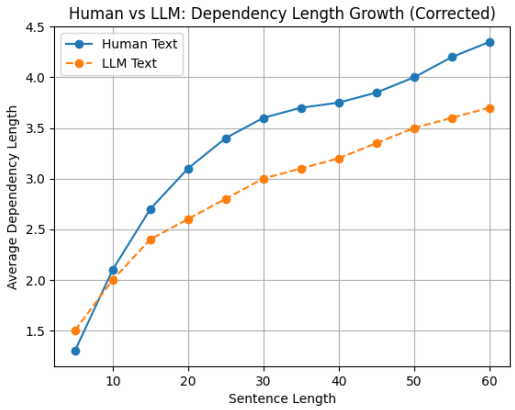

Figure 4: Human V/s LLM structure Growth trend

- Although LLM text has slightly higher average dependency length, it does not maintain the same level of structural control, resulting in flatter and less consistent growth patterns.

6.3 Dependency Length vs. Prediction Difficulty

Contrary to the initial expectation, increasing dependency length did not consistently increase surprisal. In several cases, the model maintained stable prediction confidence even under longer dependencies. Sentence-level perplexity also did not always increase and occasionally decreased for longer constructions. This suggests that long dependencies alone are not sufficient to create processing difficulty for the model.

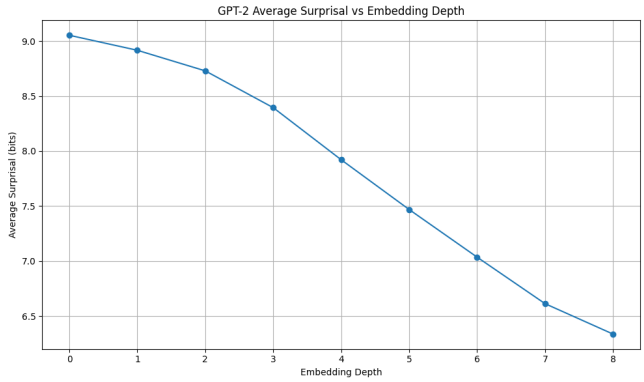

Figure 5: Average surprisal vs. embedding depth

6.4 Effect of Hierarchical Embedding

When testing sentences with increasing embedding depth, surprisal at key tokens increased in some cases, especially around structurally important words such as the main verb. However, the increase was not consistently monotonic across all embedding levels. This indicates partial sensitivity to hierarchical structure, but not a strong or stable representation of deeper syntax.

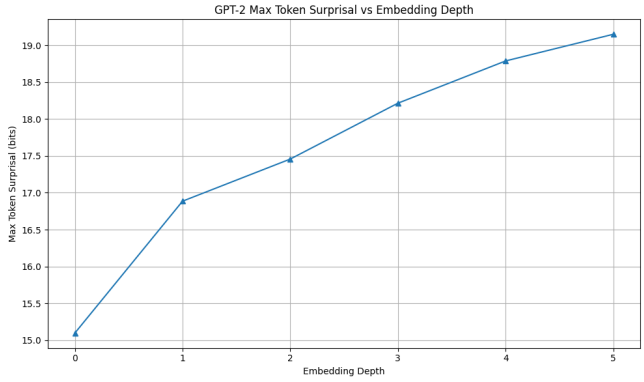

Figure 6: Maximum token surprisal vs. embedding depth

While average surprisal decreases with increasing sentence length, **maximum surprisal increases**. This indicates that prediction difficulty in LLMs is not uniformly distributed, but concentrated at specific tokens such as main verbs or clause boundaries.

6.5 Effect of Surface Form

When surface cues were reduced, prediction difficulty increased more clearly with structural complexity. Token level surprisal rise more noticeably in modified sentences than in the

original forms. This suggests that surface level regularities strongly support the model's predictions, and that deeper syntactic effects become easier to observe only when those cues are weakened.

Statistic	Surprisal	Max Surprisal	Length
Count	10000	10000	10000
Mean	9.496	16.906	15.520
Std	0.848	2.281	7.338
Min	5.867	10.648	4.000
25%	8.969	15.349	8.000
50%	9.485	16.548	13.000
75%	10.025	18.269	21.000
Max	13.499	25.602	33.000

Table 1: Summary statistics of surprisal, maximum surprisal, and sentence length.

6.6 Human vs LLM Surprisal

To compare prediction difficulty, we use surprisal as a common conceptual measure. Prior work suggests that human surprisal increases with structural complexity, especially in deeply embedded sentences.

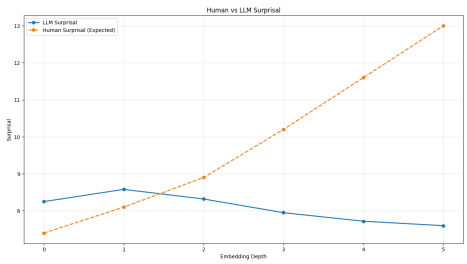

Figure 7: Human vs LLM surprisal

In contrast, our results show that LLM surprisal does not consistently increase with complexity. While spikes occur at specific tokens, overall surprisal often remains stable or decreases.

This suggests that human difficulty grows with structure, whereas LLMs rely more on contextual prediction than hierarchical constraints.

6.7 multi model analysis

The multi-model analysis shows that average surprisal does not consistently increase with embedding depth across models. Instead, surprisal tends to decrease or remain stable,

Depth	LLM Surprisal	Human Surprisal
0	8.25	7.4
1	8.58	8.1
2	8.32	8.9
3	7.95	10.2
4	7.72	11.6
5	7.60	13.0

Table 2: Comparison of surprisal trends across increasing structural complexity.

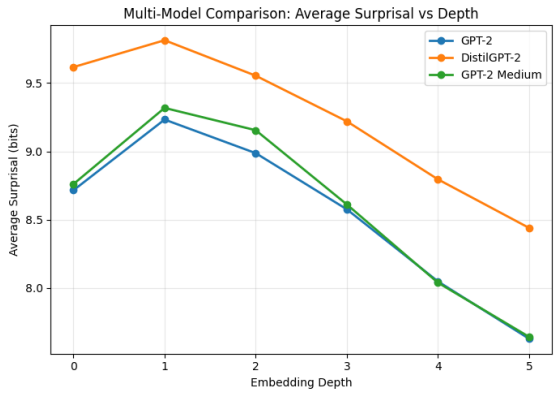

Figure 8: Average Surprisal Trends

suggesting that longer context may improve overall predictability. This pattern is consistent across GPT-2, DistilGPT-2, and GPT-2 Medium, indicating that the behavior is not model-specific. In contrast, maximum token

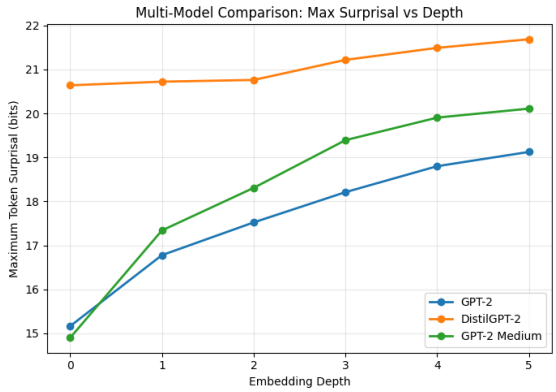

Figure 9: Max Surprisal Trends

surprisal increases steadily with depth across all models, highlighting that prediction difficulty is concentrated at specific tokens rather than uniformly distributed across the sentence.

The attention heatmap reveals that the model strongly attends to nearby tokens, with limited focus on distant dependencies. This is further supported by the observed decrease

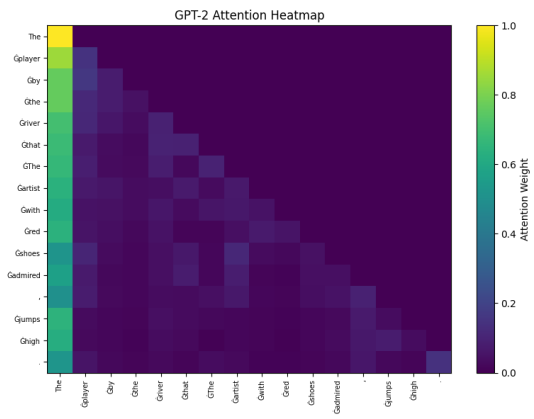

Figure 10: Attention Heatmap

in attention to earlier context as embedding depth increases, indicating that the model does not consistently maintain long-range structural relationships. Together, these results suggest that while transformer models capture some structural cues, they rely primarily on local context and do not robustly encode deeper hierarchical dependencies.

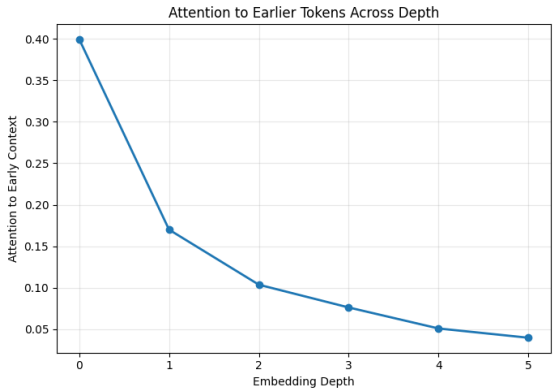

Figure 11: Attention vs Depth

6.8 Summary of Findings

- LLM-generated text exhibits relatively short dependency lengths, suggesting patterns broadly consistent with Dependency Length Minimization (DLM), similar to human language.

- Average dependency length increases with sentence length, but remains bounded, indicating that LLMs maintain structurally efficient representations even in longer sentences.

- Greater dependency length does not consistently correspond to higher prediction difficulty, suggesting a weak or non-linear relationship between structural distance and surprisal.

- Average surprisal tends to decrease with increasing sentence length, indicating that longer contexts may improve overall predictability for the model.

- In contrast, maximum surprisal increases with sentence complexity, highlighting that prediction difficulty is localized to specific tokens rather than uniformly distributed across the sentence.

- Hierarchical complexity (e.g., center embedding) affects surprisal, but the effect is partial and inconsistent, providing limited support for the hypothesis that deeper structures increase processing difficulty.

- Structural effects become more pronounced when surface-level cues are reduced, suggesting that LLMs rely heavily on lexical and contextual signals in addition to syntactic structure.

- Correlation analyses indicate a stronger relationship between sentence length and dependency length than between dependency length and surprisal, reinforcing the idea that structural complexity alone does not determine prediction difficulty.

- Overall, LLMs capture some structural regularities of human language, but their handling of complexity differs, with difficulty emerging in localized regions rather than accumulating globally.

7. DISCUSSION AND THEORETICAL IMPLICATIONS

The results suggest that transformer language models capture some structural regularities observed in human language, but do not behave like humans in a fully cognitively grounded sense. The low average dependency length in generated text indicates that LLM outputs often resemble DLM-like patterns. However, this similarity does not necessarily imply human-like memory

constraints; it may instead arise from statistical regularities present in training data.

We also expected longer dependencies to produce consistently greater prediction difficulty. That pattern did not hold reliably. The model often handled long-distance relations without a large increase in surprisal, indicating that dependency length alone does not determine model difficulty in the same way it affects human processing.

The hierarchical experiments provide a more nuanced result. Increasing embedding depth sometimes raised surprisal, especially around key structural tokens, but the effect was not strong or monotonic across all depths. This suggests that the model is somewhat sensitive to syntactic hierarchy, but does not robustly encode deeper structure.

One of the clearest findings is the role of surface form. When cues such as capitalization patterns were simplified, the effect of structural complexity became more visible. This indicates that the model depends heavily on shallow surface-level cues, and that apparent syntactic competence may partly reflect cue-driven prediction rather than abstract hierarchical understanding.

The results partially contradict this hypothesis, as increasing dependency length does not consistently lead to higher prediction difficulty in LLMs. This suggests that the relationship between structural complexity and prediction difficulty is more complex than initially expected.

8. INTERACTIVE SYSTEM

To make the analysis more interpretable and reusable, we also developed an interactive web-based interface. The system allows users to input custom sentences and inspect token-level surprisal, top-k predictions, and other model behaviors in real time. This interface complements the static graphs in the report by providing an exploratory view of how prediction changes across different sentence structures.

Such a tool is especially useful for comparing minimally different sentence forms, observing where surprisal spikes occur, and qualitatively examining whether the model responds to syntactic changes or surface modifications.

Interactive Interface: `https://cgs.outputs.muragesh.tech`

Key Takeaways

- LLMs are fundamentally predictive systems that rely heavily on contextual patterns rather than explicit structural reasoning.
- Longer context often improves overall predictability, as reflected by decreasing average surprisal with sentence length.
- Prediction difficulty is not uniformly distributed; instead, it is concentrated at specific tokens such as main verbs or structurally critical points.
- Dependency length alone is not a reliable indicator of prediction difficulty in LLMs, unlike in human language processing.
- Hierarchical structure is only partially captured, suggesting that LLMs do not consistently encode deep syntactic relationships.
- Surface-level cues play a significant role in model behavior, and reducing these cues reveals underlying structural weaknesses.
- Interactive experiments (via the provided web interface) demonstrate that small changes in phrasing or structure can significantly affect token-level surprisal.
- Overall, LLMs behave more like advanced pattern recognizers than systems with human-like memory constraints or structural understanding.

Scope and Future Work

The current study focuses on controlled sentence constructions to systematically analyze the effect of structural complexity. While this enables precise manipulation of syntactic properties, extending the analysis to more diverse natural language data could provide additional insights.

The experiments are conducted using a single pretrained model. Evaluating multiple transformer architectures may help determine whether the observed trends generalize across models.

Future work can further explore fine-grained token-level behavior and incorporate more linguistically varied datasets to better understand how prediction difficulty emerges in different contexts.

9. CONCLUSION

This project examined whether transformer-based language models exhibit human-like structural constraints by combining dependency-length analysis with surprisal-based prediction experiments. The results show that LLM-generated text often resembles human language in preferring shorter dependencies, but increasing dependency length does not consistently increase prediction difficulty. Hierarchical complexity affects prediction only partially, while surface cues play a major supporting role. Overall, the findings suggest that LLMs capture some outward patterns of human language structure, but their behavior is more strongly driven by learned statistical regularities than by human-like memory constraints.

These findings suggest that the structural patterns observed in transformer language models are not the result of human like cognitive constraints, but rather emerge as by-products of large scale statistical learning.

10. APPENDIX

Appendix A: Implementation Details

The project was implemented in Python. The codebase includes modules for dependency parsing, dependency-length computation, surprisal and perplexity calculation, data processing, and visualization. Standard NLP libraries were used for parsing, while transformer-based inference was performed using a pretrained GPT-2 model.

Appendix B: Code Repository

The complete implementation of the project is publicly available and includes all datasets, preprocessing pipelines, analysis scripts, and visualization modules used in this study. This repository enables full reproducibility of the results and provides additional implementation details beyond the scope of the main report.

GitHub Link:
`https://github.com/Muragesh-24/CGS-410-COURSE-PROJECT`

Appendix C: Interactive Web Interfaces

To complement the static analysis presented in this report, we developed interactive web interfaces that allow dynamic exploration of model behavior and experimental results.

Main Interactive Interface: This interface enables users to input custom sentences and analyze their surprisal, attention patterns, and structural properties in real time. It is designed to provide an intuitive understanding of how transformer-based models process language.

Interactive Interface: `https://cgs.outputs.muragesh.tech`

Additional Output Visualizations: This page contains a comprehensive collection of generated plots, including dependency length distributions, surprisal trends, and structural comparisons across different experimental settings. These visualizations supplement the figures included in the main report.

Outputs display: `https://cgs.outputs.muragesh.tech/outputs.html`

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